

Graphpad Prism 5.0

解析 (1)

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王盟



共享资源

GraphPad 官方网站:

<http://www.graphpad.com> 可下载相关软件, 免费试用。

应用简介视频 (英语) :

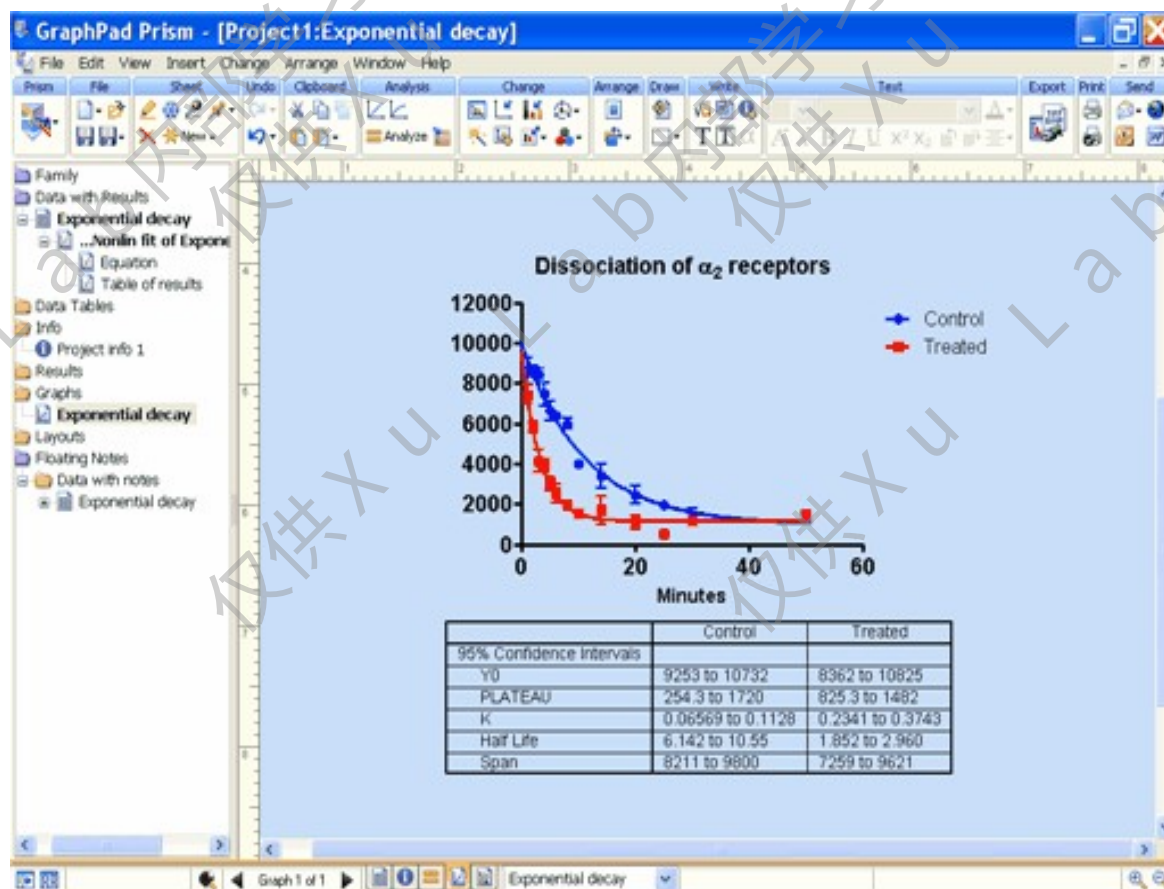
<http://www.graphpad.com/prism/prism5movieweb.html>

软件简介

What is GraphPad Prism?

是一款集数据分析和作图于一体的数据处理软件，它可以直接输入原始数据，自动进行基本的生物统计，如计算标准差、标准误和 P 值等，同时产生高质量的科学图表。

图
是这款软件的
核心。



软件自带的学习资料

- **Movies**

最舒服的一种方式。

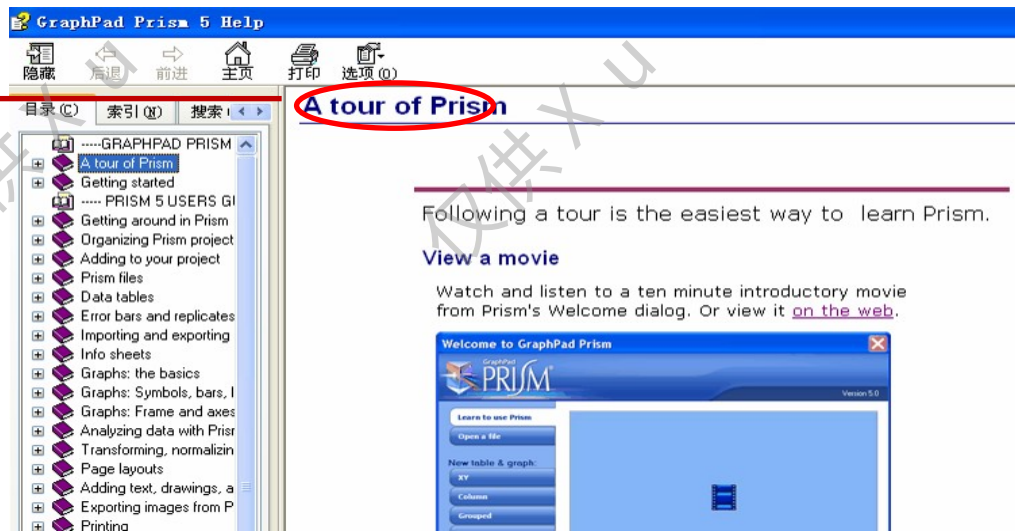
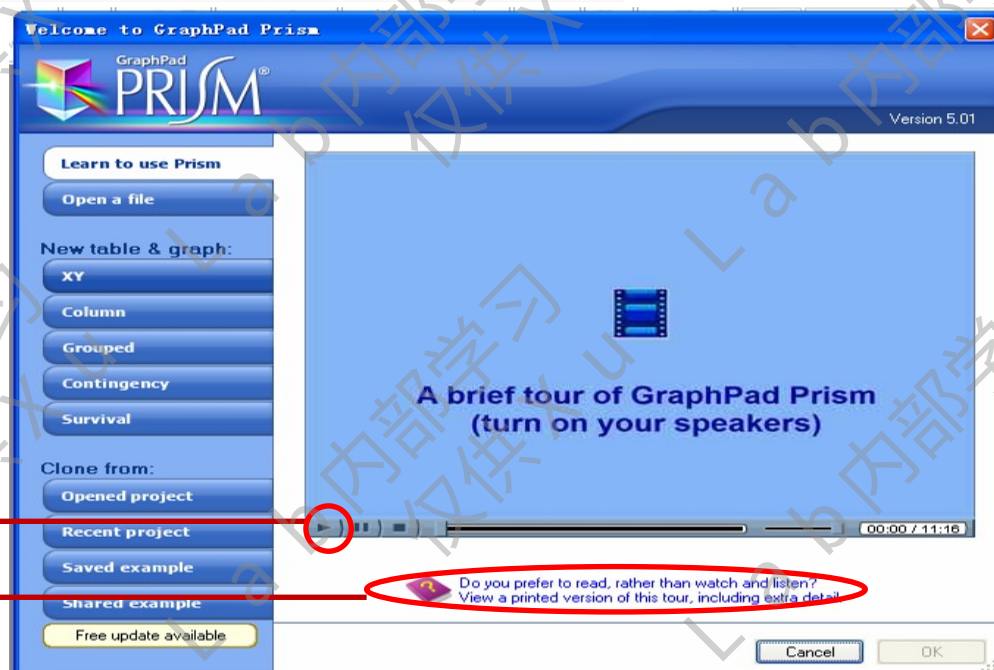
- **详细使用说明**

最详细，内容多需花大力气
研究学习。

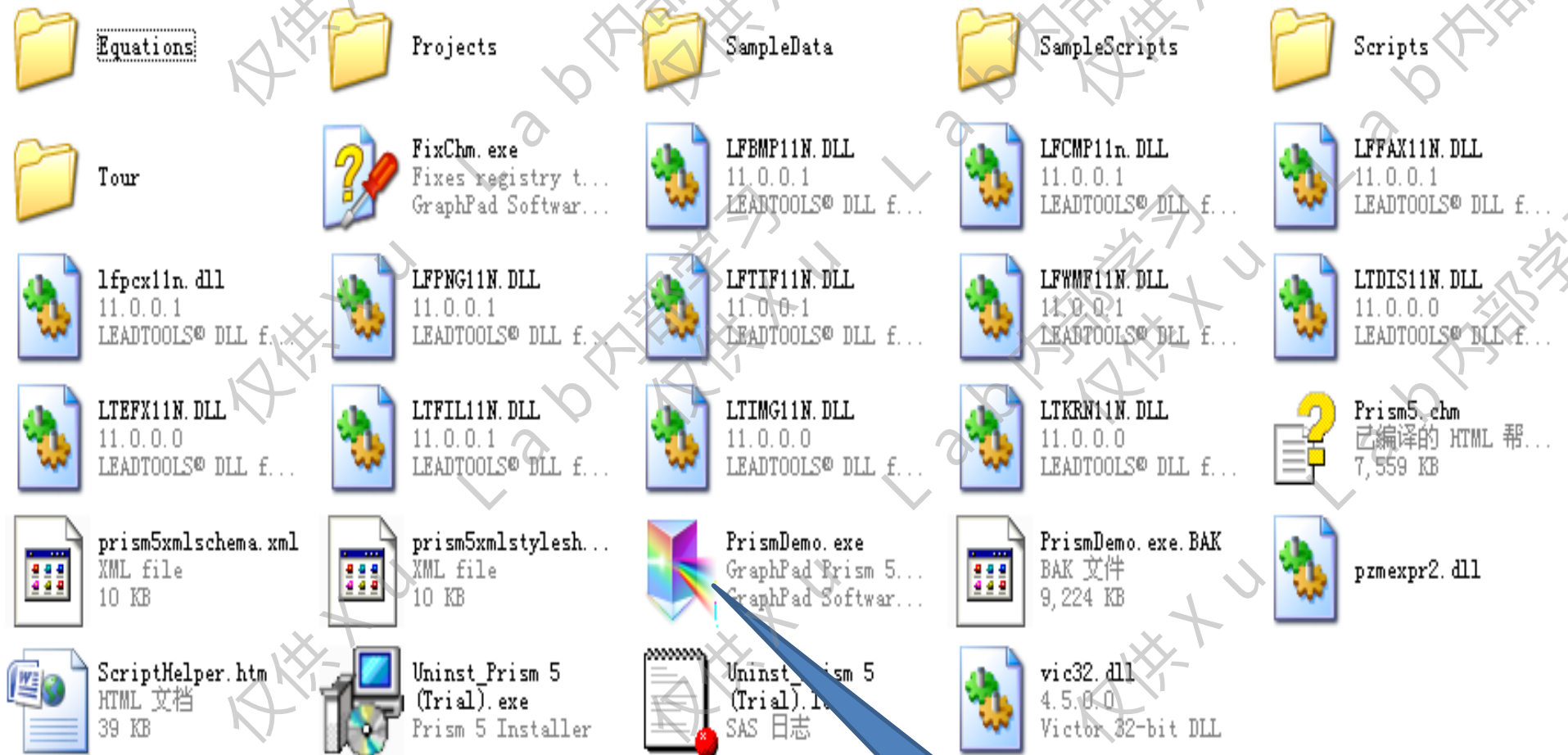
- **快速入门手册**

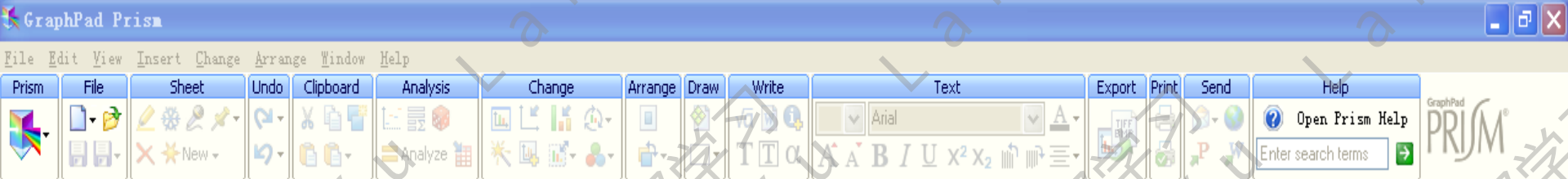
快速掌握基本技能知识
和新特点。

边用边学

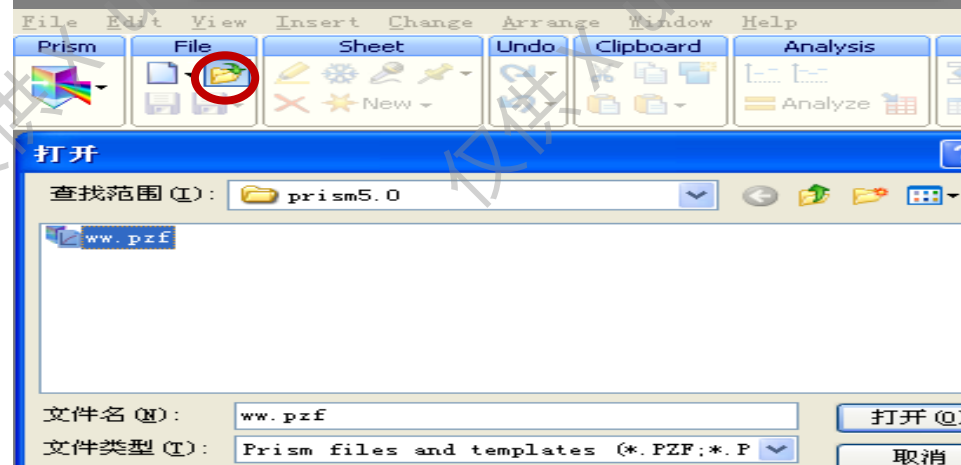
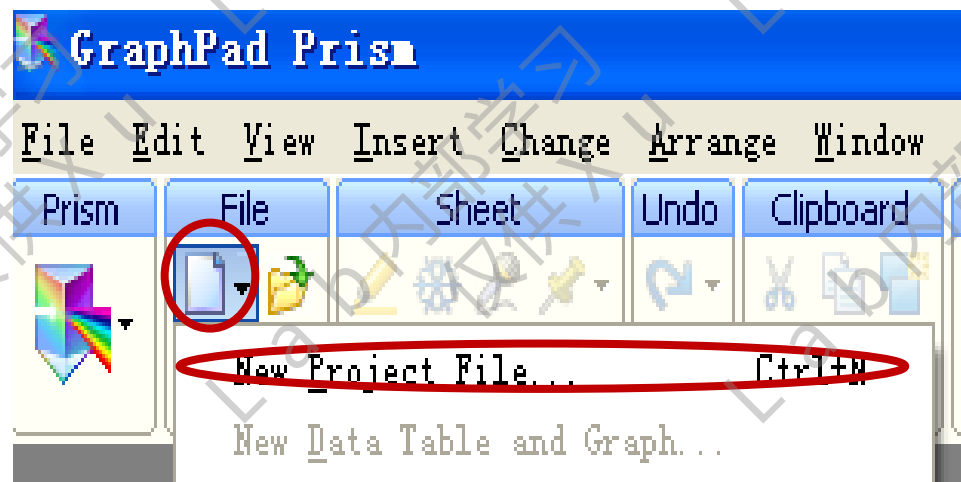
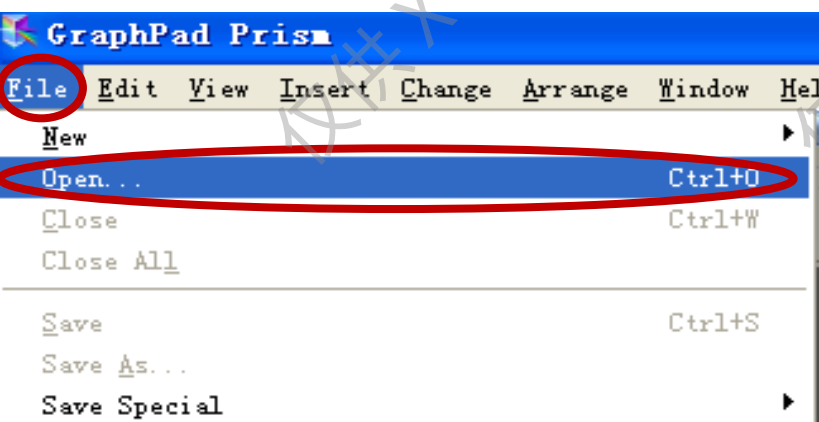
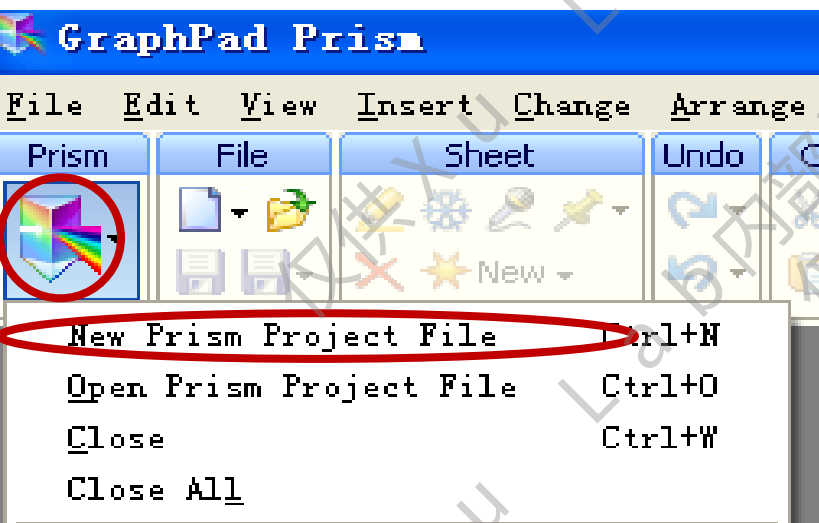


一切从头开始





新建文件及打开已有文件的方法



Column 分析模块

Learn to use Prism

Open a file

New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Available analyses

- t test (one-sample, paired and unpaired)
- Wilcoxon
- Column statistics (including normality tests)
- Correlation matrix
- One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
- Kruskal-Wallis
- Friedman

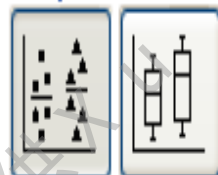
Organization of data table

Sample data

Start with an empty data table

Use sample data

Choose a graph



Selected graph: Scatter plot

Graphing replicates or error bars

How is a Column data table organized?

How is a Column data table organized?

Making a column bar graph

Frequency distribution and histogram

t test - Unpaired

t test - Paired

t test - One sample

One-way ANOVA - Ordinary

One-way ANOVA - Repeated measures

ROC curve

Bland-Altman method comparison

Correlation matrix

Metanalysis (Forest) plot

以列变量做柱状图

以列变量做频数分布或柱状图

非配对 t 检验

配对 t 检验

单样本 t 检验

单因素方差分析

重复测量单因素方差分析

感受性曲线

两种方法测量一致性检验

相关矩阵

Met 分析

New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Recent project

Saved example

Shared example

Organization of data table

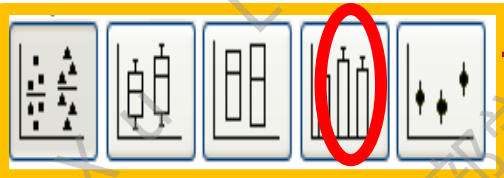
Sample data

☐ Start with an empty data table

☒ Use sample data

t test - Unpaired

Choose a graph



Selected graph: Scatter plot, vertical

Graphing replicates or error bars

Plot:

Mean with SEM

Mean

Mean with SD

Mean with SEM

Mean with 95% CI

Mean with range

Geometric mean

Geometric mean with 95% CI

Median

Median with range

No line or error bar

Cancel

Create

作图选择，上下两排图
XY 坐标互换

配对分析时的图

图及其上的“工”
字符表示含义

均数及标准差

均数及标准误

均数及 95% 可信区间

均数及观察值范围

几何均数

几何均数及 95% 可信区
间

中位数

图上无“工”字符

非配对 t 检验

Welcome to GraphPad Prism

GraphPad
PRISM[®]

Version 5.00 (Trial), 三月 12, 2007

Learn to use Prism

Open a file

New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Recent project

Saved example

Shared example

Available analyses

- t test (one-sample, paired and unpaired)
- Mann-Whitney
- Wilcoxon
- Column statistics (including normality tests)
- Correlation matrix
- One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
- Kruskal-Wallis
- Friedman

[Organization of data table](#)

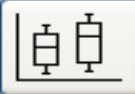
Sample data

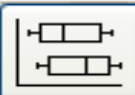
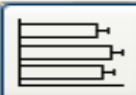
☐ Start with an empty data table

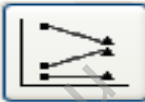
☒ Use sample data

t test - Unpaired

Choose a graph



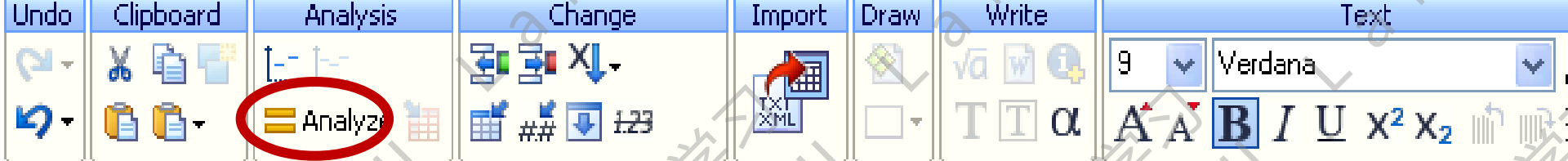
Selected graph: **Column bar graph, vertical**

Graphing replicates or error bars

Plot: **Mean with SD**

Choose how to calculate and plot error bars from raw data. To directly enter mean and error values, create a Grouped table, and use only one row.

Cancel **Create**



	A	B	C	D	E	F	G	H
	Male	Female	Title	Title	Title	Title	Title	Title
	Y	Y	Y	Y	Y	Y	Y	Y

1	54.	43
2	23	34
3	45	65
4	54	77
5	45	46
6		65

☒

How the data are organized

The two columns define two groups. Note that, unlike many statistics programs, Prism does not define groups using a grouping variable. Instead, the groups are defined by columns.

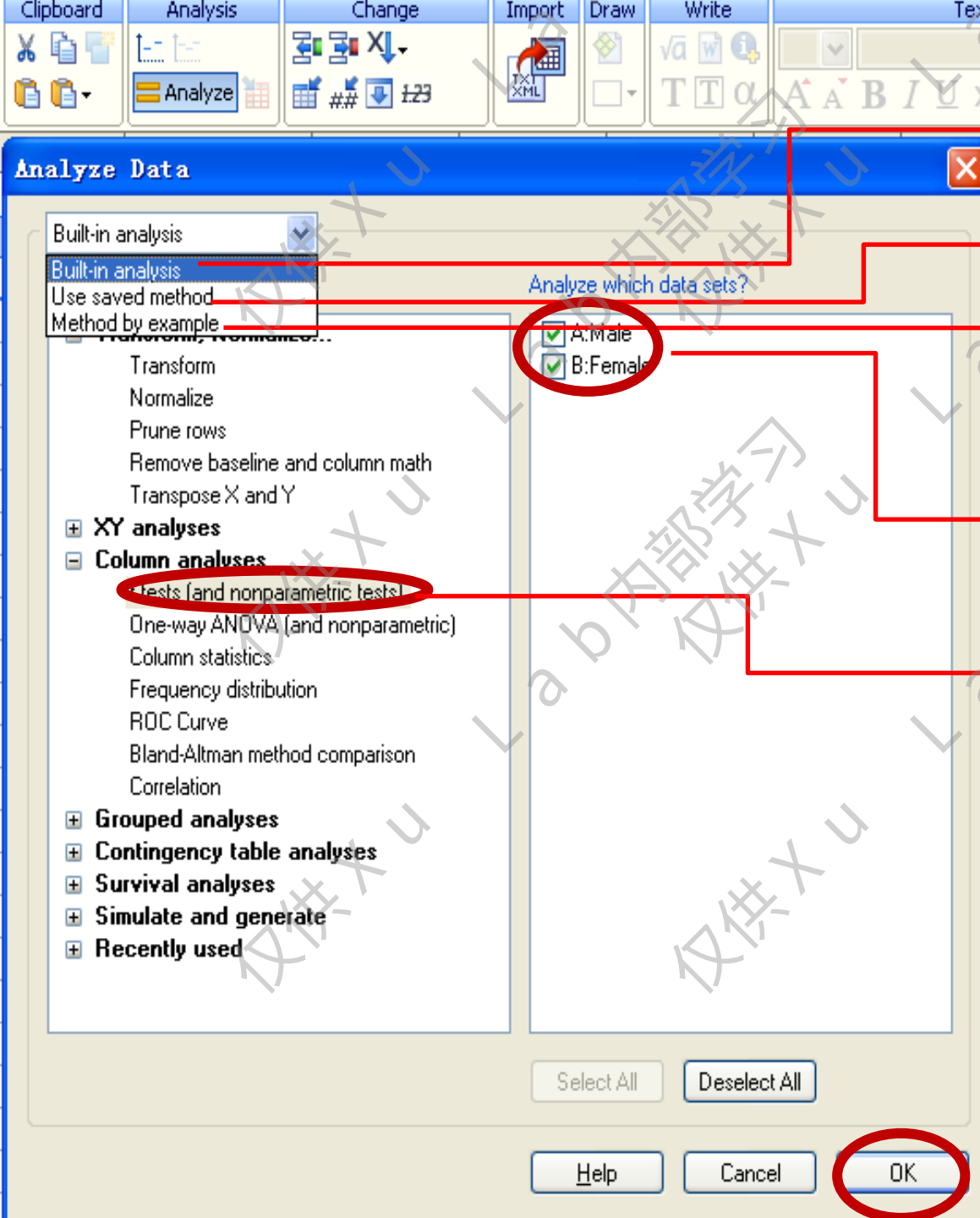
The goals

- To determine if the differences between the two group means is greater than you'd expect to see by chance.
- To determine the 95% confidence interval for the difference between the two means.

How to perform an unpaired t test

Click Analyze, choose t test from the list of column analyses, then choose an unpaired t test on the dialog. Click the link below for detailed instructions.

[Step-by-step instructions for performing an unpaired t test](#)



使用软件列出的分析方法

使用自己存储的分析方法

使用举例的分析方法

选中要分析的变量

选中要分析的变量

Parameters: t Tests (and Nonparametric Te...

Choose Test

You may either choose a test by checking the three option boxes, or you may choose a test by name below.

☐ Paired test. Values in each row represent paired observations.

☐ Nonparametric test. Don't assume Gaussian distributions.

☐ Welch's correction. Don't assume equal variances.

Test Name: Unpaired t test

Options

P values:

Confidence Interval:

- Mann-Whitney test
- Wilcoxon matched pairs test
- Unpaired t test with Welch's correction
- Paired t test
- Unpaired t test

Significant digits

Show 4 significant digits

Output

☐ Create a table of descriptive statistics for each column

Learn

Cancel

OK

资料符合正态分布时的配对检验

资料不符合正态分布时的非参数检验

资料方差不齐时 Welch's 校准后独立样本均数比较的秩和检验

非参检验，两独立样本检验

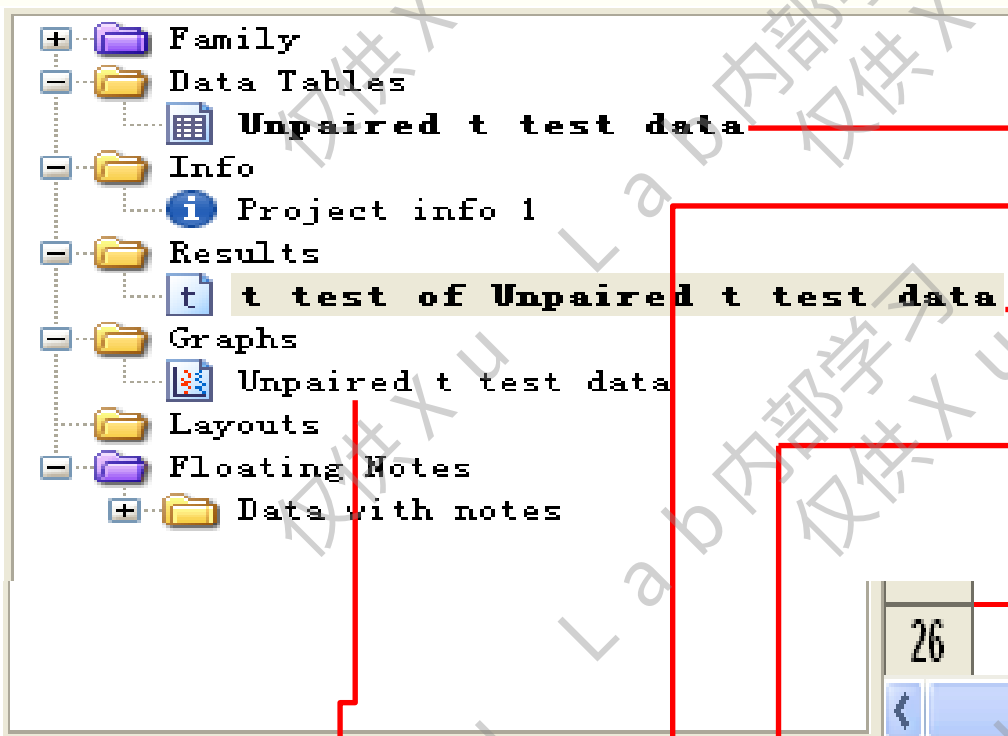
非参检验，两相关样本检验（配对、配伍）

用 Welch's 校准的两样本非配对 t 检验

正态分布资料配对样本均数比较 t 检验

正态分布资料非配对样本均数比较 t 检验

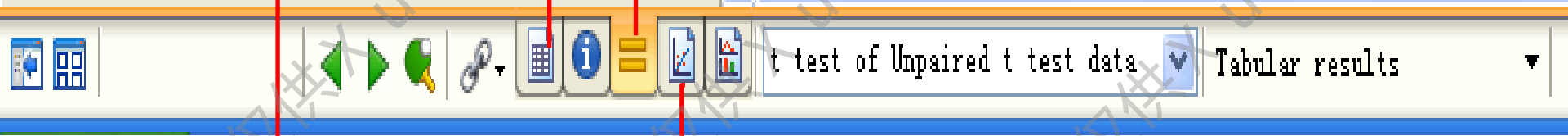
对每列资料进行描述统计分析



数据表

数据表

统计图





t test Tabular results

t 检验结果

1	Table Analyzed	Unpaired t test data
2	Column A	Male
3	vs	vs
4	Column B	Female
5		
6	Unpaired t test	
7	P value	0.2613
8	P value summary	ns
9	Are means signif. different? (P < 0.05)	No
10	One- or two-tailed P value?	Two-tailed
11	t, df	t=1.199 df=9
12		
13	How big is the difference?	
14	Mean ± SEM of column A	44.20 ?5.669 N=5
15	Mean ± SEM of column B	55.00 ?6.708 N=6
16	Difference between means	-10.80 ?9.010
17	95% confidence interval	-31.18 to 9.582
18	R squared	0.1377
19		
20	F test to compare variances	
21	F,DFn, Dfd	1.680, 5, 4
22	P value	0.6354
23	P value summary	ns
24	Are variances significantly different?	No

统计方法

P
值

统计结果判定

t
值

自由度

两组方差齐 性检验

两组方差齐



t test

Column statistics

A

B

列变量特征

Male

Female

Y

Y

1 Number of values

5

6

2 Minimum

23.00

34.00

3 25% Percentile

34.00

40.75

4 Median

45.00

55.50

5 75% Percentile

54.00

68.00

6 Maximum

54.00

77.00

7

8 Mean

44.20

55.00

9 Std. Deviation

12.68

16.43

10 Std. Error

5.669

6.708

11

12 Lower 95% CI

28.46

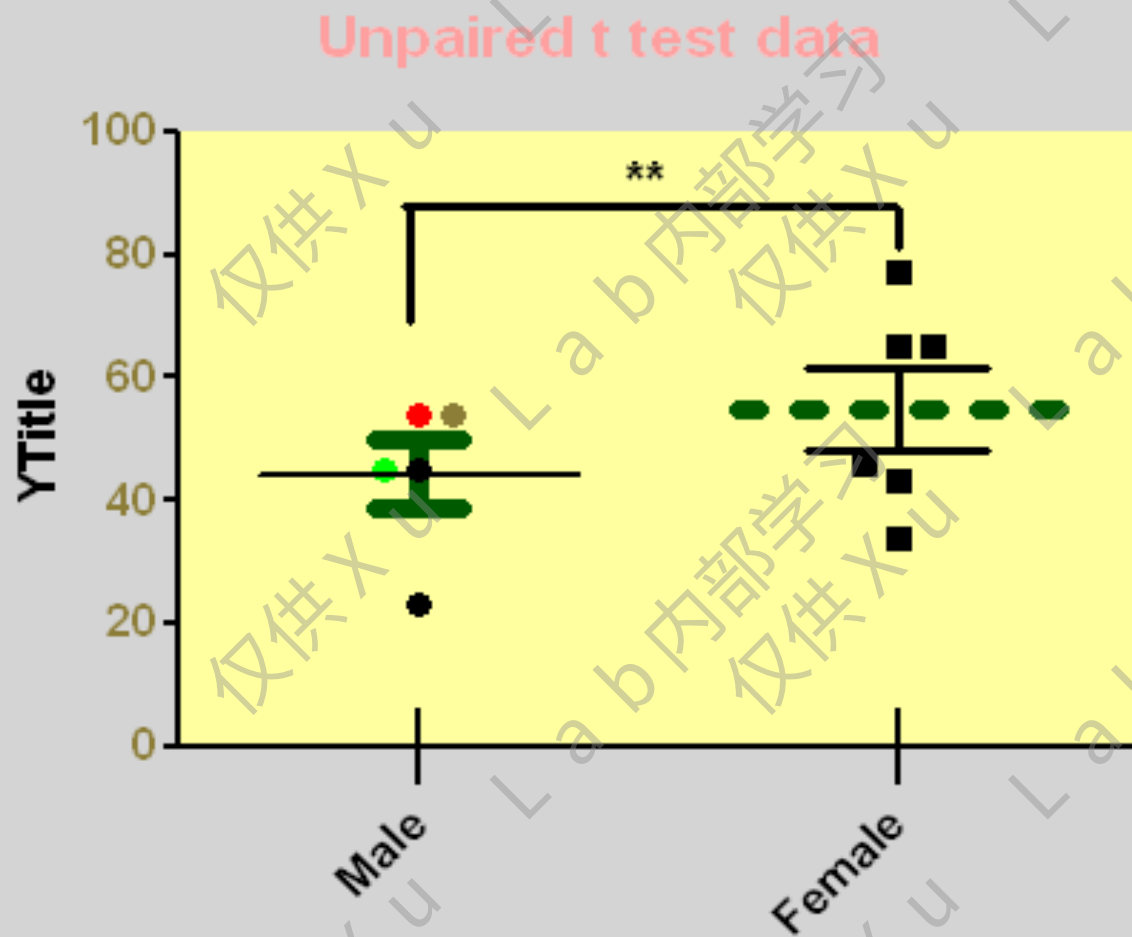
37.76

13 Upper 95% CI

59.94

72.24

均数、标准差、
标准误



男性	女性
54	43
23	34
45	65
54	77
45	46
	65

Figure 2. Proportions of T cell receptor (TCR) T cells and NK cells in healthy donors (HDs), asymptomatic carriers (AsCs) of hepatitis B virus, patients with low-grade chronic hepatitis B (CHB-L), and patients with moderate or severe chronic hepatitis B (CHB-M/S). The distribution statistics

Frame and Origin

X axis

Left Y axis

Right Y axis

Titles & Fonts

Origin

Set origin: Automatically

Y intersects the X axis at X= 1.0

X intersects the Y axis at Y= 0.0

Shape, Size and Position

Size: Wide

Width (Length of X axis): 7.62 cm

Height (Length of Y axis): 5.08 cm

Distance of Y axis from left edge: 12.75 cm

Distance of X axis from bottom edge: 11.99 cm

Axes and Colors

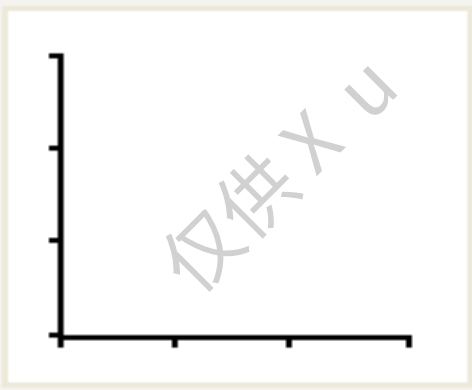
Thickness of axes: 1 pt

Color of axes:

Color of plotting area: Clear

Page background: Clear

Frame and Grid Line



Frame style: No frame

Hide axes: None

☐ Show Scale Bar

Major grid: None

Color: Auto

Thickness: 1 pt

Style:

Minor grid: None

Color: Auto

Thickness: 1/2 pt

Style:

Format Graph



Appearance

Data Sets on Graph

Graph Settings

Data set: Unpaired t test data:Male



All

Style

Appearance: Scatter dot plot



Line at: Mean with SEM



Symbols

Color:



Shape:



Size:



Border color:



Border thickness:

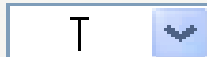


Error bars

Color:



Style:



Dir:



Thickness:



Lines

Color:



Thickness:



Line goes:



Style:



Pattern:



Additional options

Plot on: ☒ Left Y axis

☐ Right Y axis

☐ Show legend

☐ Revert legend to column title

☐ Label each point with its row title

Help

Cancel

OK

配对 t 检验

Welcome to GraphPad Prism



Version 5.00 (Trial), 三月 12, 2007

Learn to use Prism

Open a file

New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Recent project

Saved example

Shared example

Available analyses

- t test (one-sample, paired and unpaired)
- Mann-Whitney
- Wilcoxon
- Column statistics (including normality tests)
- Correlation matrix
- One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
- Kruskal-Wallis
- Friedman

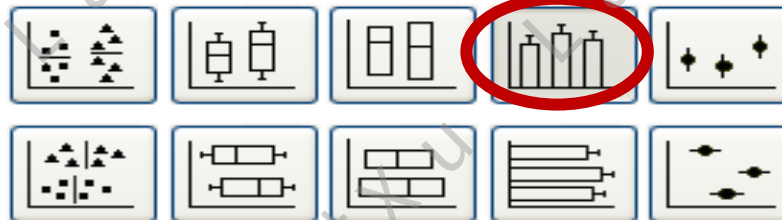
[Organization of data table](#)

Sample data

- ☐ Start with an empty data table
- ☒ Use sample data

t test - Paired

Choose a graph



Selected graph: **Column bar graph, vertical**

Graphing replicates or error bars

Plot: **Mean with SEM**

Choose how to calculate and plot error bars from raw data. To directly enter mean and error values, create a Grouped table, and use only one row.

Cancel

Create

Parameters: t Tests (and Nonparametric Te...

Analyze Data

Built-in analysis ▾

Which analysis?

- ☒ **Transform, Normalize...**
 - Transform
 - Normalize
 - Prune rows
 - Remove baseline and column math
 - Transpose X and Y
- ☒ **XY analyses**
- ☒ **Column analyses**
 - t tests (and nonparametric tests)**
 - One-way ANOVA (and nonparametric)
 - Column statistics
 - Frequency distribution
 - ROC Curve
 - Bland-Altman method comparison
 - Correlation
- ☒ **Grouped analyses**
- ☒ **Contingency table analyses**
- ☒ **Survival analyses**
- ☒ **Simulate and generate**
- ☒ **Recently used**

Analyze which data sets?

- ☒ A: Before
- ☒ B: After

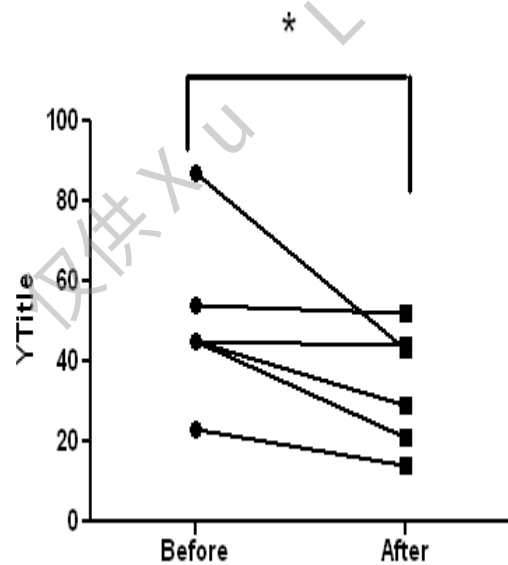
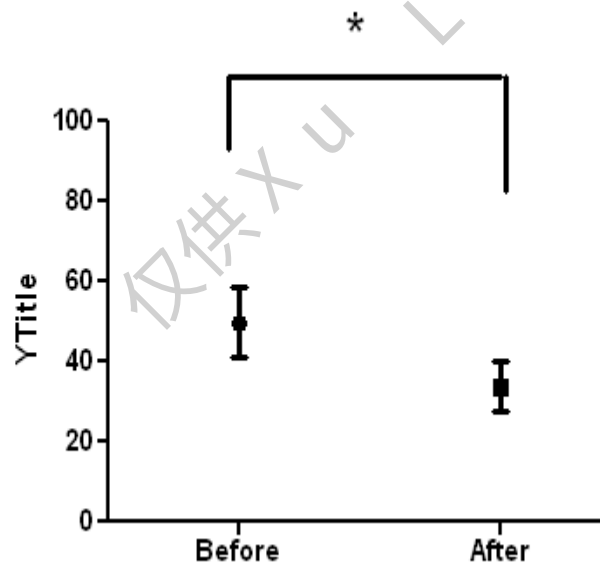
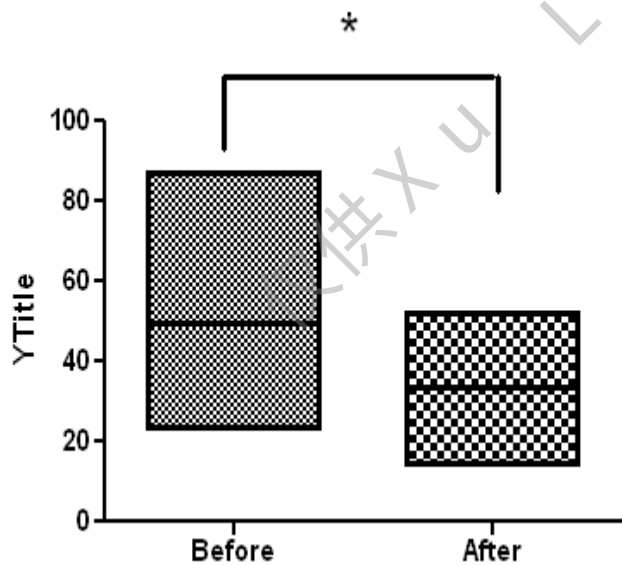
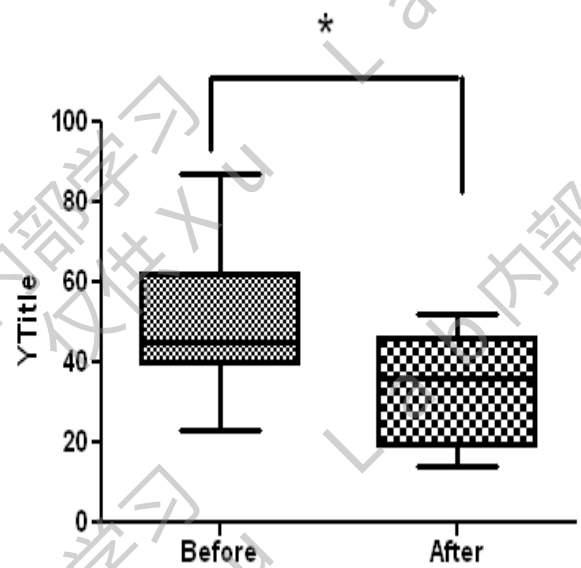
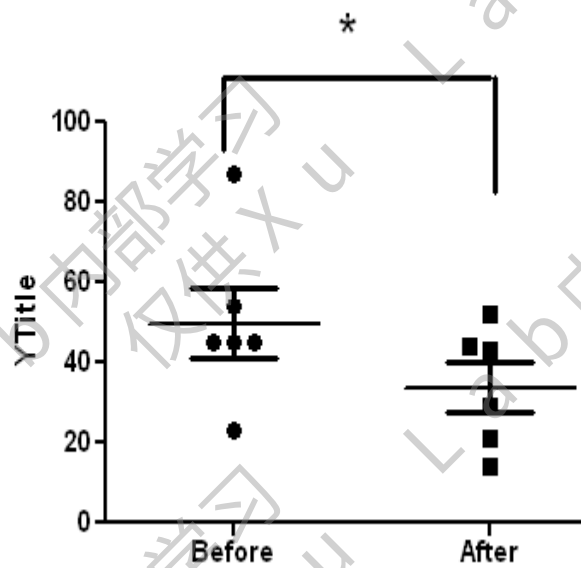
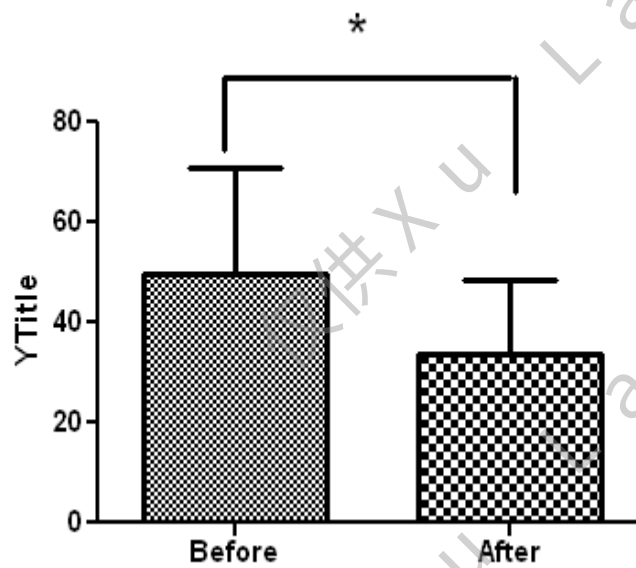
Select All Deselect All

Help Cancel OK

t test		
1	Table Analyzed	Paired t test data
2	Column A	Before
3	vs	vs
4	Column B	After
5		
6	Paired t test	
7	P value	0.0606
8	P value summary	ns
9	Are means signif. different? ($P < 0.05$)	No
10	One- or two-tailed P value?	Two-tailed
11	t, df	t=2.414 df=5
12	Number of pairs	6
13		
14	How big is the difference?	
15	Mean of differences	16.00
16	95% confidence interval	-1.041 to 33.04
17	R squared	0.5382
18		
19	How effective was the pairing?	
20	Correlation coefficient (r)	0.6350
21	P Value (one tailed)	0.0878
22	P value summary	ns
23	Was the pairing significantly effective?	No
24		

$P > 0.05$

差异无统计学意义



New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Recent project

Saved example

Organization of data table

Sample data

☐ Start with an empty data table☒ Use sample data

One-way ANOVA - Ordinary

Choose a graph



Selected graph: Scatter plot, vertical

Grouping replicates or error bars

Plot: Mean with SD

Undo | Clipboard | Analysis | Change | Import

Analyze

	A	B	C
	Control	Treated	Treated+Antagonist
	Y	Y	Y
1	54.	87	45
2	23	98	39
3	45	64	51
4	54	77	49
5	45	89	50
6	47		55
7			
8			

Analyze Data

Built-in analysis

Which analysis?

☒ Transform, Normalize...

Transform
Normalize
Prune rows
Remove baseline and column math
Transpose X and Y

☒ XY analyses☒ Column analyses

t tests (and nonparametric tests)
One-way ANOVA (and nonparametric)
Column statistics
Frequency distribution
ROC Curve
Bland-Altman method comparison
Correlation

☒ Grouped analyses☒ Contingency table analyses☒ Survival analyses☒ Simulate and generate☒ Recently used

Analyze which data sets?

☒ A: Control
☒ B: Treated
☒ C: Treated+Antagonist

Select All

Deselect All

Help

Cancel

OK

Parameters: One-Way ANOVA (and Nonparametric)



Choose test

You may either choose a test by checking the two option boxes, or you may choose a test by name below.

☐ Repeated measures test. Values in each row represent matched observations.

☐ Nonparametric test. Don't assume Gaussian distributions.

Test name: One-way analysis of variance

Post test

Test name: One-way analysis of variance

Kruskal-Wallis test

Friedman test

Repeated Measures ANOVA

Significance level. Alpha = 0.05 (95% confidence intervals)

Control column: A:Control

Select...

Significant digits

Show 4 significant digits

Output

☐ Create a table of descriptive statistics for each column

Learn

Cancel

OK

重复测量，单行各列重复

非参数检验

非参检验：多样本中位数
比较是否有差异

非参检验：双向方差分析检验
多样本秩和是否有差

单因素方差分析：多样本均数
比较是否有差异

重复测量资料的方差分析

Parameters: One-Way ANOVA (and Nonparametric)



Choose test

You may either choose a test by checking the two option boxes, or you may choose a test by name below.

☐ Repeated measures test. Values in each row represent matched observations.

☐ Nonparametric test. Don't assume Gaussian distributions.

Test name: One-way analysis of variance

Post test

Test name: No Post Test

No Post Test

Tukey: Compare all pairs of columns.

Newman-Keuls: Compare all pairs of columns

Bonferroni: Compare all pairs of columns.

Bonferroni: Compare selected pairs of columns.

Dunnnett: Compare all columns vs. control column

Test for linear trend between mean & col. number

Significant

Show 4

Output

☐ Create a table of descriptive statistics for each column

Learn

Cancel

OK

无组间多重比较的单因素方差分析

各组间比较的单
因素方差分析

各组间比较的单
因素方差分析

各组间比较的单
因素方差分析

三方法差别
不大

选定两组间比较的单
因素方差分析

方差不齐时，各组与同一对
照组比较单因素方差分析

线性趋势检验

Table Analyzed	One-way ANOVA data		
One-way analysis of variance			
P value	P<0.0001		
P value summary	***		
Are means signif. different? (P < 0.05)	Yes		
Number of groups	3		
F	22.57		
R squared	0.7633		
Bartlett's test for equal variances			
Bartlett's statistic (corrected)	2.986		
P value	0.2247		
P value summary	ns		
Do the variances differ signif. (P < 0.05)	No		
ANOVA Table	SS	df	MS
Treatment (between columns)	4760	2	2380
Residual (within columns)	1476	14	105.4
Total	6236	16	
Tukey's Multiple Comparison Test	Mean Diff.	q	Significant? P < 0.05?
Control vs Treated	-38.33	8.719	Yes
Control vs Treated+Antagonist	-3.500	0.8349	No
Treated vs Treated+Antagonist	34.83	7.923	Yes

P<0.01

差异有统计学意义

统计量 F=22.57

Bartlett's 方差齐性检验

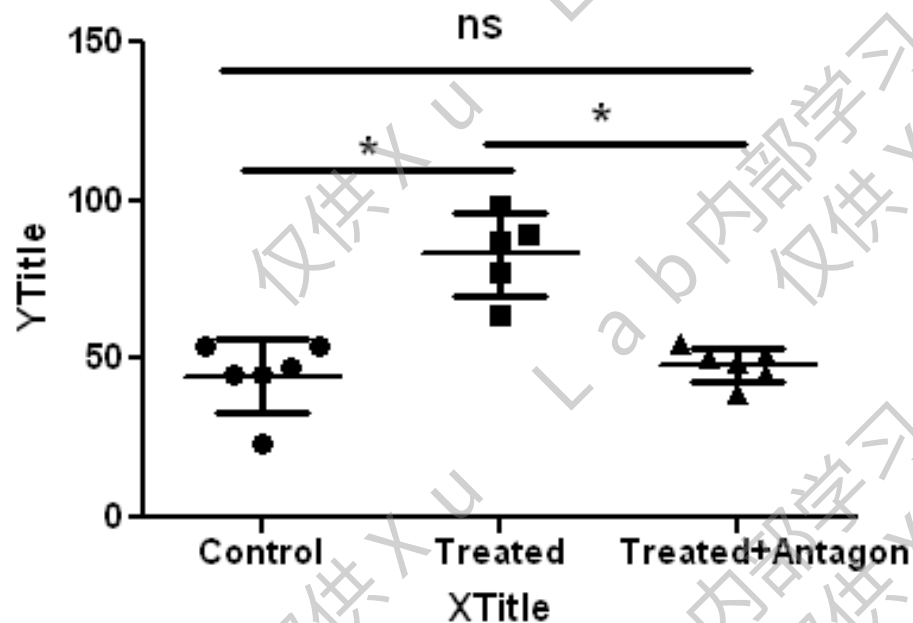
方差齐性检验 p>0.05

Bartlett's 检验方差齐

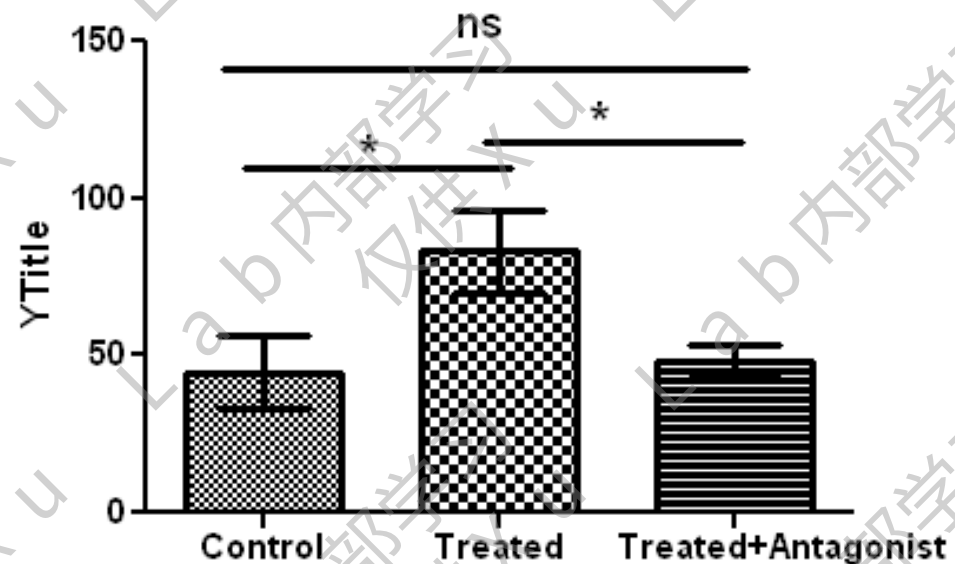
Tukey's 多重比较差异有统计学意义

Control 与 Treated+Ant 组比较差异无统计学意义

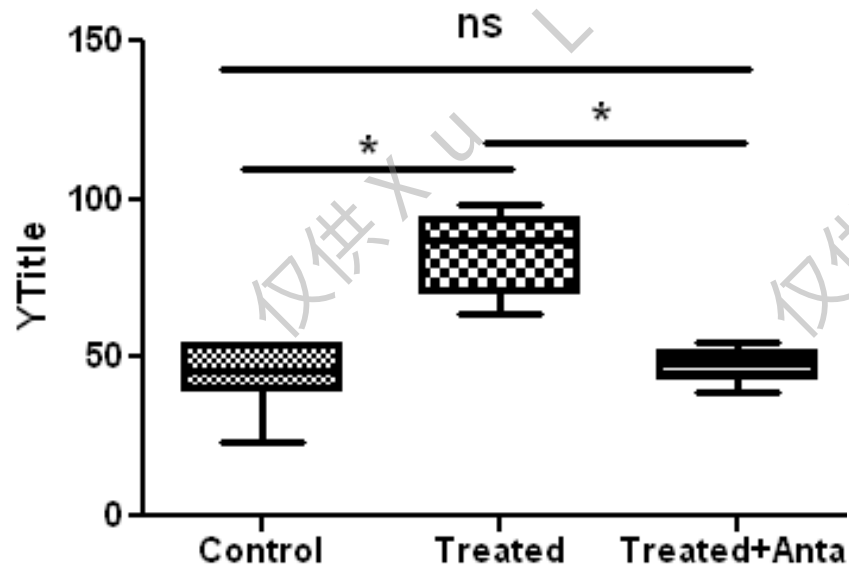
One-way ANOVA data



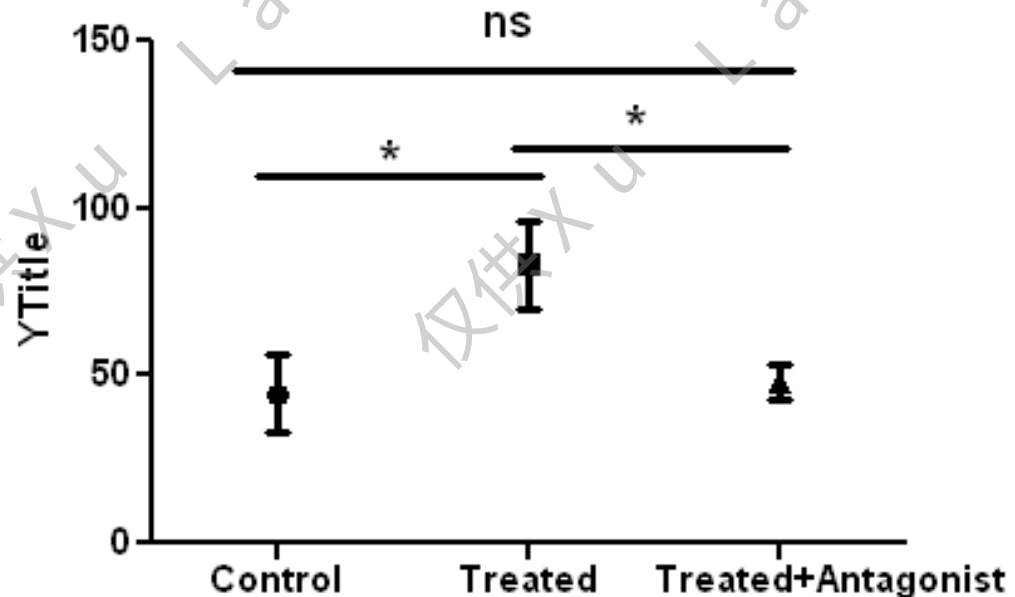
One-way ANOVA data



One-way ANOVA data



One-way ANOVA data



One-way anova - Repeated measures (单因素方差分析 - 重复测量)

Welcome to GraphPad Prism



Version 5.00 (Trial), 三月 12, 2007

Learn to use Prism

Open a file

New table & graph:

XY

Column

Grouped

Contingency

Survival

Clone from:

Opened project

Recent project

Saved example

Shared example

Available analyses

- t test (one-sample, paired and unpaired)
- Mann-Whitney
- Wilcoxon
- Column statistics (including normality tests)
- Correlation matrix
- One-way ANOVA (followed by Tukey, Dunnett, Newman-Keuls or Bonferroni post tests)
- Kruskal-Wallis
- Friedman

[Organization of data table](#)

Sample data

☐ Start with an empty data table

☒ Use sample data

One-way ANOVA - Repeated measures

Choose a graph



Selected graph: **Column bar graph, vertical**

Graphing replicates or error bars

Plot:

Mean with SD

Choose how to calculate and plot error bars from raw data. To directly enter mean and error values, create a Grouped table, and use only one row.

Cancel

Create

Undo	Clipboard	Analysis	Change	Import	Draw
		 Analyze			
	A	B	C	D	
	Control	Treatment 1	Treatment 2	Treatment 3	
	Y	Y	Y	Y	
1	54.	43	78	111	
2	23	34	65	99	
3	45	65	99	78	
4	54	77	79	90	
5	45	46	87	95	

Analyze Data

Built-in analysis

Which analysis?

- Transform, Normalize...
 - Transform
 - Normalize
 - Prune rows
 - Remove baseline and column math
 - Transpose X and Y
- XY analyses
- Column analyses
 - t tests (and nonparametric tests)
 - One-way ANOVA (and nonparametric)**
 - Column statistics
 - Frequency distribution
 - ROC Curve
 - Bland-Altman method comparison
 - Correlation
- Grouped analyses
- Contingency table analyses
- Survival analyses
- Simulate and generate
- Recently used

Analyze which data sets?

- ☒ A: Control
- ☒ B: Treatment 1
- ☒ C: Treatment 2
- ☒ D: Treatment 3

Select All Deselect All

Help Cancel OK

Parameters: One-Way ANOVA (and Nonparametric)

Choose test

You may either choose a test by checking the two option boxes, or you may choose a test by name below.

☒ Repeated measures test. Values in each row represent matched observations.

☐ Nonparametric test. Don't assume Gaussian distributions.

Test name: Repeated Measures ANOVA

Post test

Test name: No Post Test

Significant Control

- No Post Test
- Tukey: Compare all pairs of columns.
- Newman-Keuls: Compare all pairs of columns.
- Bonferroni: Compare all pairs of columns.**
- Bonferroni: Compare selected pairs of columns.
- Dunnnett: Compare all columns vs. control column.
- Test for linear trend between mean & col. number**

Show 4

Output

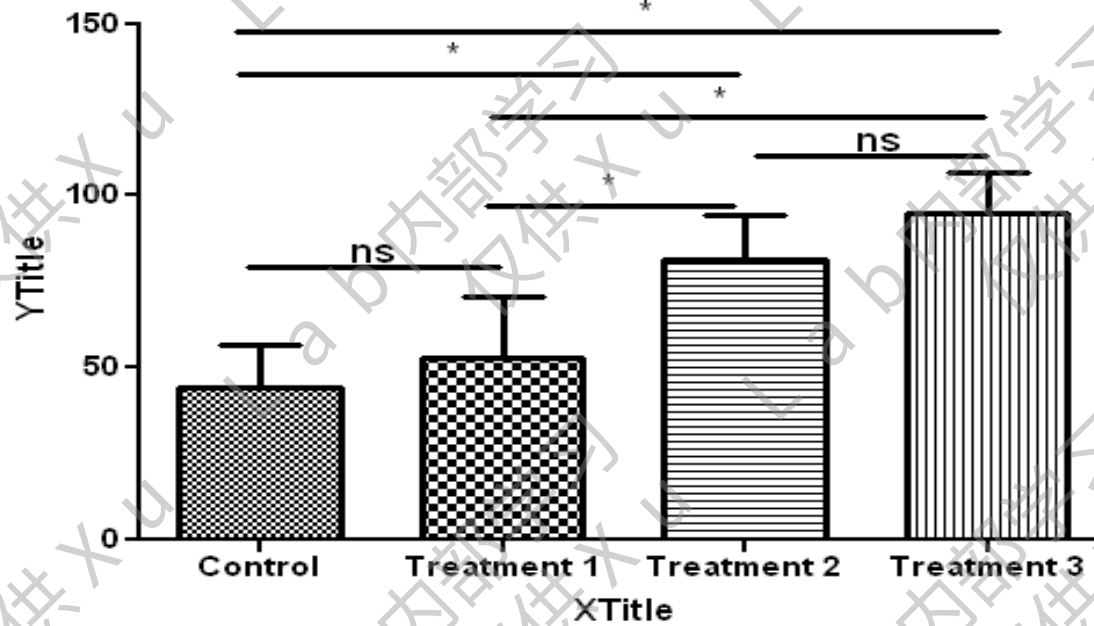
☐ Create a table of descriptive statistics for each column

Learn Cancel OK

3	Repeated Measures ANOVA			
4	P value	0.0002		
5	P value summary	***		
6	Are means signif. different? (P < 0.05)	Yes		
7	Number of groups	4		
8	F	15.76		
9	R squared	0.7976		
10				
11	Was the pairing significantly effective?			
12	R squared	0.08253		
13	F	1.333		
14	P value	0.3134		
15	P value summary	ns		
16	Is there significant matching? (P < 0.05)	No		
17				
18	ANOVA Table	SS	df	MS
19	Treatment (between columns)	8417	3	2806
20	Individual (between rows)	949.3	4	237.3
21	Residual (random)	2136	12	178.0
22	Total	11500	19	
23				
24	Bonferroni's Multiple Comparison Test	Mean Diff.	t	Significant? P < 0.05?
25	Control vs Treatment 1	-8.800	1.043	No
26	Control vs Treatment 2	-37.40	4.432	Yes
27	Control vs Treatment 3	-50.40	5.973	Yes
28	Treatment 1 vs Treatment 2	-28.60	3.390	Yes
29	Treatment 1 vs Treatment 3	-41.60	4.930	Yes
30	Treatment 2 vs Treatment 3	-13.00	1.541	No

1	Table Analyzed	Repeated measures one-way ANOVA data		
2				
3	Repeated Measures ANOVA			
4	P value	0.0002		
5	P value summary	***		
6	Are means signif. different? ($P < 0.05$)	Yes		
7	Number of groups	4		
8	F	15.76		
9	R squared	0.7976		
10				
11	Was the pairing significantly effective?			
12	R squared	0.08253		
13	F	1.333		
14	P value	0.3134		
15	P value summary	ns		
16	Is there significant matching? ($P < 0.05$)	No		
17				
18	ANOVA Table	SS	df	MS
19	Treatment (between columns)	8417	3	2806
20	Individual (between rows)	949.3	4	237.3
21	Residual (random)	2136	12	178.0
22	Total	11500	19	
23				
24	Post test for linear trend			
25	Slope	8.990		
26	R squared	0.7026		
27	P value	$P < 0.0001$		
28	P value summary	***		
29	Is linear trend significant ($P < 0.05$)?	Yes		

Repeated measures one-way ANOVA data



Repeated measures one-way ANOVA data

